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Predictive analytics in customer behavior: Anticipating trends and preferences

Hamed GhorbanTanhaei, Payam Boozary *, Sogand Sheykhan, Maryam Rabiee, Farzam Rahmani, Iman Hosseini

Amirkabir University of Technology, Department of Management, Science & Technology, Iran

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ABSTRACT

In order to effectively manage their customers, businesses need to thoroughly analyze the costs and advantages associated with various alternative expenditures and investments and determine the most effective way to allocate resources to marketing and sales activities over time. Those in charge of making decisions will reap the benefits of decision support models that estimate the value of the customer portfolio and tie expenses to customers' purchasing behavior. In the current work, various machine learning algorithms such as Decision Tree (DT), Random Forest (RT), Logistic Regression (LR), Support Vector Machines (SVM), and gradient boosting are used to predict customer behavior. The evaluation criteria considered in the work include precision, recall, F1-Score, and ROC-AUC. The accuracy values obtained for DT, RT, LR, SVM, and gradient boosting are 0.787, 0.806, 0.826, 0.826, and 0.823, respectively. The results emphasize RT and LR's good performance, while the values of 0.620, 1, 0.766, and 0.878 for the precision, recall, F1-score, and ROC-AUC score outperform the rest. The novelty of this work lies in employing a comprehensive set of machine learning algorithms to predict customer behavior, with a particular emphasis on the superior performance of RF and LR models, as demonstrated by their high precision, recall, F1-score, and ROC-AUC values.

1. Introduction

1.1. Background

In today's business environment, firms must comprehend and forecast client behavior to enhance their marketing and sales tactics [22]. This study aims to utilize machine learning algorithms to monitor and forecast customer behavior, thereby aiding organizations in allocating resources and making informed decisions [7,41]. The new adaptive management attempts to be sensitive to local needs and to enable cooperation among diverse stakeholders [24]. This is in contrast to the work done in the past on scientific adaptive management [13]. In the dynamic landscape of modern business, understanding and anticipating customer behavior is paramount to success. The advent of predictive analytics has revolutionized how companies approach this challenge, offering a powerful toolkit to decipher patterns, trends, and preferences from vast datasets. By harnessing historical data and deploying sophisticated algorithms, businesses can comprehend the intricacies of customer behavior and forecast future actions [4]. This proactive approach enables

* Corresponding author.
E-mail address: Payam.boozary@aut.ac.ir (P. Boozary).

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organizations to tailor their strategies, from personalized marketing initiatives and adaptive customer journeys to dynamic pricing models and effective churn prevention [36]. In this era of data-driven decision-making, predictive analytics serves as a guiding force, empowering businesses to stay ahead of the curve, deliver enhanced customer experiences, and ultimately thrive in an ever-evolving marketplace [26]. This introduction sets the stage for exploring the multifaceted applications of predictive analytics in deciphering and shaping customer behavior trends [28]. The proliferation of the internet has led to the digitalization of data, which has created opportunities for using big data. By leaving traces of what customers see, what they read, their involvement and behavior, and judgment about their interests and preferences, digital data in large numbers provides a large amount of data that can be mined for learning experiences [20]. This data can provide insights into customers' preferences and interests. The value of big data lies in the outcomes of analyses and predictions, as well as the actions that should be taken based on those analyses and predictions [6]. Data utilization, statistical algorithms, and machine learning techniques are the components that make up predictive analytics. Predictive analytics aims to identify potential patterns, events, and behaviors in the future based on historical data [39].

Customer behavior refers to individuals' actions and decision-making processes when they purchase goods or services [22]. It encompasses various factors, including how customers select, purchase, use, and dispose of products and the emotional, mental, and behavioral responses that precede or follow these actions. Understanding customer behavior is critical for businesses as it provides insights into customer needs, preferences, and buying patterns. This knowledge allows companies to tailor their marketing strategies, improve product offerings, and enhance customer satisfaction and loyalty. Analyzing customer behavior involves studying demographic data, purchasing history, and feedback, which helps predict future buying tendencies and identify potential areas for business growth [33]. It is becoming more common for businesses to shift their focus from product-centered to customer-centered [9]. In order to develop long-term, lucrative relationships with individual consumers in a highly competitive global economy, it is essential to have a comprehensive grasp of customers' behavior. For this reason, predicting the behavior of customers is an essential component of analytical customer relationship management (CRM). To summarize, the primary components of customer behavior prediction are the prediction of customer turnover and the forecast of client purchasing activities [44]. Regarding loyalty programs, predicting customer attrition is an important aspect. This initiative aims to identify consumers who are likely to switch at least some of their purchases from a particular firm and to assist businesses in developing intervention tactics that will encourage these customers to remain loyal to the company. To summarize, the primary components of customer behavior prediction are the prediction of customer turnover and the forecast of client purchasing activities [5].

Regarding loyalty programs, predicting customer attrition is an important aspect [31]. This initiative aims to identify consumers who are likely to switch at least some of their purchases from a particular firm and to assist businesses in developing intervention tactics that will encourage these customers to remain loyal to the company. The use of information technology in today's workplaces is becoming increasingly ubiquitous [16]. A significant expansion has been seen in information technology and its applications. Any company serious about enhancing its prospects in the industry would unquestionably maintain a close eye on the perspectives of its customers and the trends that are always developing. These businesses are using a vast array of technological concepts, tools, and procedures in order to get a better understanding of their prospective employees. Better decisions may be made by doing more in-depth analyses of data and facts. It is a widely held belief among professionals in business intelligence [17]. Among the most difficult challenges mobile telecom carriers face, customer churn is one of the most significant factors influencing revenue and client base. The effectiveness of retention efforts is dependent not solely on the precision of identifying future churners but also, and this is of equal significance, on the time if forecasting occurs [2]. Leveraging predictive analytics to understand customer behavior must be considered in today's competitive business landscape. This sophisticated analytical approach allows companies to move beyond reactive strategies and embrace a proactive stance, anticipating trends and preferences before they fully manifest. Businesses can optimize operations, enhance customer satisfaction, and drive sustainable growth by accurately predicting customer behavior. This predictive foresight enables targeted marketing efforts, personalized customer experiences, and the ability to swiftly adapt to changing market dynamics. Moreover, it aids in resource allocation, ensuring efforts focus on areas with the highest potential impact. Ultimately, the importance of predictive analytics in customer behavior lies in its capacity to empower businesses with actionable insights, fostering strategic decision-making, fostering customer loyalty, and maintaining a competitive edge in an environment where adaptability and foresight are key determinants of success [3].

1.2. Objectives

The primary objective of this research is to assess the effectiveness of different machine learning models, such as DT, RF, LR, SVM, and Gradient Boosting, in forecasting customer behavior. The evaluation will be based on precision, recall, F1-score, and ROC-AUC performance metrics. The conceptual framework for this study is grounded in the theory of customer relationship management (CRM) and the application of data-driven decision-making. Existing literature underscores the importance of CRM in understanding customer lifetime value and improving customer retention rates. Previous studies have used machine learning techniques to predict customer churn, purchase behavior, and segmentation. For instance, research by Ngai et al. highlighted the efficacy of data mining techniques in CRM [32], while recent studies by Lariviere and Van den Poel [23] emphasized the use of predictive models to enhance marketing strategies. However, this study addresses a gap in comprehensive comparative analyses of multiple machine learning models on a unified dataset. The relevance of this research lies in its potential to provide actionable insights for businesses seeking to refine their customer engagement strategies. By employing a diverse set of machine learning algorithms, this study identifies the most effective models for customer behavior prediction and contributes to the broader field of data-driven marketing. The research findings are expected to benefit both academia and industry by offering a robust evaluation of machine learning techniques in CRM applications, thus advancing knowledge in the field and providing a basis for future research. This study presents a sophisticated approach

to customer behavior prediction, utilizing diverse machine learning algorithms, including DT, RT, LR, SVM, and gradient boosting. Through a meticulous evaluation process employing precision, recall, F1-Score, and ROC-AUC metrics, our findings reveal the exceptional performance of RF and LR models, underscoring their significance in efficiently allocating resources and making informed decisions for customer management in the dynamic business landscape. The main contributions are as follows:

- The primary objective of this study is to assess and compare the performance of different machine learning models in predicting customer behavior. Specifically, the study aims to:
- Identify the most accurate machine learning models for predicting customer behavior.
- Evaluate the performance of these models using key metrics such as accuracy, precision, recall, F1-score, and ROC-AUC.
- Provide insights into the strengths and weaknesses of each model in the context of CRM applications.

2. Literature review

The significance of this work lies in its profound implications for the field of customer management and business strategy [19]. By employing diverse machine learning algorithms, including LR and RF, to predict customer behavior, the study contributes to a deeper understanding of the intricacies involved in consumer decision-making. These models' accuracy and robust performance underscore their practical applicability in optimizing resource allocation, refining marketing strategies, and enhancing overall customer relationship management [12]. The insights gained from this research provide a valuable framework for decision-makers to tailor their approaches based on predicted customer actions and pave the way for more data-driven and effective business practices. In an era where customer-centricity is paramount, the findings of this study offer a roadmap for organizations to stay competitive and responsive to dynamic market trends, ultimately fostering sustainable growth and success in today's rapidly evolving business landscape. This has also been explored in prior studies. For instance, by operationalizing the link between green marketing's impact on consumer attitudes and the mediating role of the marketing mix towards green products, Chou et al. [10] validated the suggested methodology regarding consumers' willingness to be environmentally friendly in the Taiwanese context. The model was constructed using structural equation modeling (SEM) and data gathered from 977 online consumers. According to the results, attitudes towards green products substantially and indirectly influenced green consumption intention. This study examined the relationship quality between social media marketing activity (SMMA) and customer experience (CX) to influence consumer behavioral outcomes such as intention to purchase, remain devoted, and participate. Using SmartPLS 3, the 413 online questionnaire surveys were measured and analyzed. The findings indicate that both CX and SMMA have a substantial effect on the character of customer relationships, which positively influences customer behavior. Wibowo et al. [43] offered guidance to enterprises regarding the necessity for SNS marketing content to adhere to the SMMA and CX dimensions to accomplish marketing objectives and foster sustainable performance. Momin et al. [30] empirically compared an artificial neural network and several conventional classification algorithms—K-nearest Neighbours, Naive Bayes, Random Forest, Decision Tree, and LR —to forecast customer attrition on the IBM Telco Customer attrition dataset. The models were evaluated and contrasted in terms of their precision in forecasting customer attrition. The ANN model obtained an accuracy score of 82.83% on validation data, which improved over the 79.86%. Wen et al. [42] indicated a customer behavior recognition (CBR) method for achieving flexibility by combining CB and primitives. A primitive was a unit that described the motion of an object or the relationships between multiple objects. The amalgamation of various primitives could define a specific CB. Due to the fact that primitives can be repurposed to define a vast array of distinct CBs, the proposed method is adaptable to target CB modifications in retail stores. The authors utilized their own laboratory dataset and the publicly available MERL dataset in the experiments. The primitive combination was modified to accommodate the variations in target CBs across distinct datasets. The proposed method thus obtained satisfactory recognition accuracy and flexibility. This study introduces novelty by comprehensively exploring the predictive capabilities of various machine learning algorithms, including LR and RF, in customer behavior analysis. Unlike previous works, our approach considers diverse models and evaluates their performance using precision, recall, F1-Score, and ROC-AUC metrics. This departure from singular model analyses enriches the depth of insights into customer behavior prediction. Many prior studies have been limited by their focus on individual algorithms or a restricted set of evaluation criteria, overlooking the potential benefits of ensemble methods and a broader range of performance metrics. Our work addresses these limitations, providing a more holistic and nuanced understanding of the strengths and weaknesses of different models in customer management. This approach enhances the applicability and reliability of the findings, contributing to the advancement of methodologies for predictive analytics in business and marketing.

Furthermore, Abbasi Siar et al. modeled consumer impulse buying behavior using Agent-Based Modeling to analyze the effects of discounts and swarm behavior on sales, concluding that a swarm-discount approach increased sales more effectively than traditional discounts [1]. Bigdelou et al. [8] determined that the benefits and motivations of using e-commerce positively affected customer experience and consumer information, with pleasure and timely use serving as mediators, based on a survey of 384 randomly selected Digi Kala Co. customers. Li et al. [27] conceptualized a new framework using the Antifragility Analysis Algorithm (AAA) for assessing digitalization strategies in agriculture, demonstrating that IoT is the most antifragile strategy for small farms in northern Iran based on six key future indicators, thereby aiding decision-makers in managing complexity and uncertainty. Kaveh et al. [18] predicted and ranked customer attraction factors for Mellat Insurance Company in Shiraz from 2019 to 2021 using a system dynamics model and network data envelopment analysis, identifying the most efficient factors and their interactions to enhance customer orientation and competitive advantage. Zheng et al. [45] constructed single-channel and dual-channel reverse supply chain models to analyze the impact of channel competition and consumer behavior on e-platform recycling, finding that differentiated pricing benefits recyclers by increasing WEEE recycling, while e-platforms prefer single or unified dual-channel models depending on consumer preference and disposal revenue, offering insights for effective channel management and pricing strategies. Khodaverdian et al. [21] introduced a

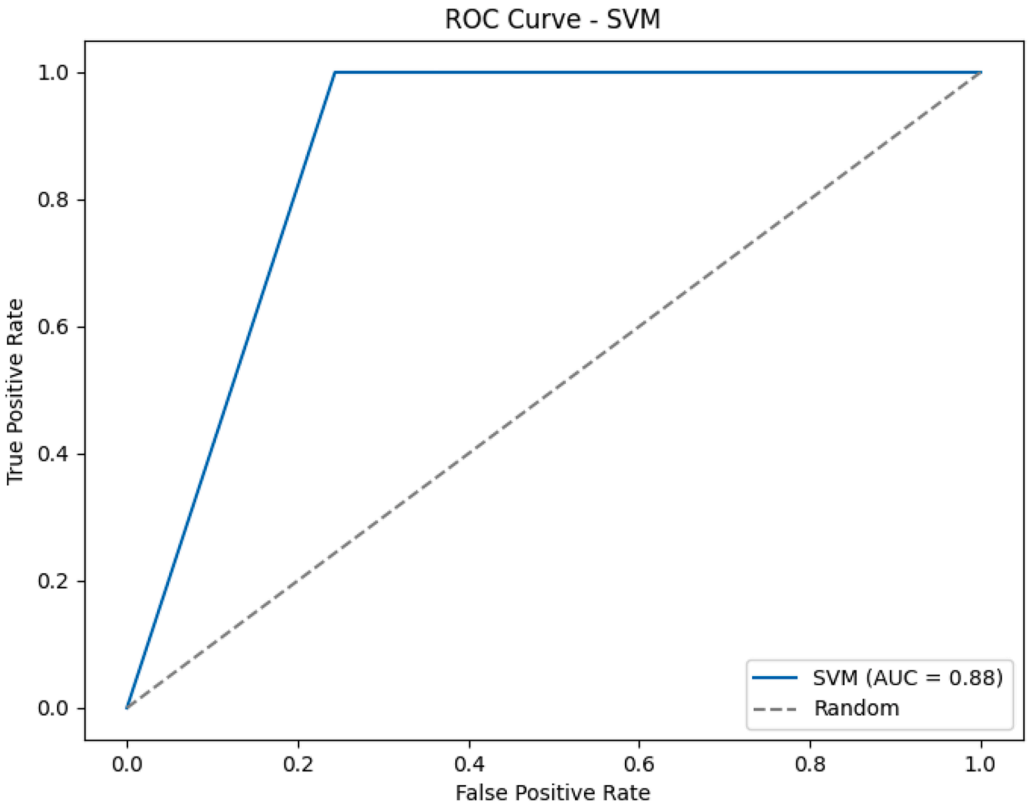

Fig. 1. The performance of SVM based on true positive rate and false positive rate.

hybrid model combining Convolutional Neural Network (CNN) and Gated Recurrent Unit (GRU) to classify virtual machines in Microsoft's Azure cloud service as Delay-Sensitive or Delay-Insensitive, addressing unbalanced sample distribution with the Synthetic Minority Oversampling Technique (SMOTE), and achieving a classification accuracy of 95.18%, thereby improving VM selection for migration and resource efficiency in cloud data centers. Supriyati and Wiyatno reviewed the literature on the SMEs service industry. They identified several methods—KANO, Servqual, IPA, and QFD—that can measure, analyze, and evaluate service quality and customer satisfaction, providing significant insights for SMEs to identify business weaknesses and prioritize improvements for continued development [38]. Rasinojehdehi and Azizi [37] addressed the pressing issue of escalating insurance costs by focusing on predicting customer claim amounts using data from Iran Insurance Company. Utilizing multiple linear regression and IBM SPSS MODELER software, the research analyzed attributes such as vehicle owner age, car type, and claim history to forecast claim amounts accurately. By providing valuable insights for insurance industry stakeholders, this approach enabled informed decision-making to mitigate financial risks and optimize operations. Hoque et al. [14] introduced Customer-Centric Market Basket Analysis (CC-MBA), integrating consumer segmentation through RFM analysis and K-means clustering, demonstrating its superiority in deriving more relevant association rules and streamlining analytical processes for businesses to improve decision-making and marketing strategies.

This study introduces novelty by comprehensively exploring the predictive capabilities of various machine learning algorithms, including LR and RF, in customer behavior analysis. Unlike previous works, our approach considers diverse models and evaluates their performance using precision, recall, F1-Score, and ROC-AUC metrics. This departure from singular model analyses enriches the depth of insights into customer behavior prediction. Many prior studies have been limited by their focus on individual algorithms or a restricted set of evaluation criteria, overlooking the potential benefits of ensemble methods and a broader range of performance metrics. Our work addresses these limitations, providing a more holistic and nuanced understanding of the strengths and weaknesses of different models in customer management. This approach enhances the applicability and reliability of the findings, contributing to the advancement of methodologies for predictive analytics in business and marketing.

3. Methodology

3.1. ML model selection

This study employed a comprehensive methodology to predict customer behavior using diverse machine learning algorithms. The process began with data preprocessing, where raw data was cleaned and standardized, and features were engineered to ensure model compatibility. Subsequently, the dataset was divided into training and testing sets to accurately assess the models' performance. Five

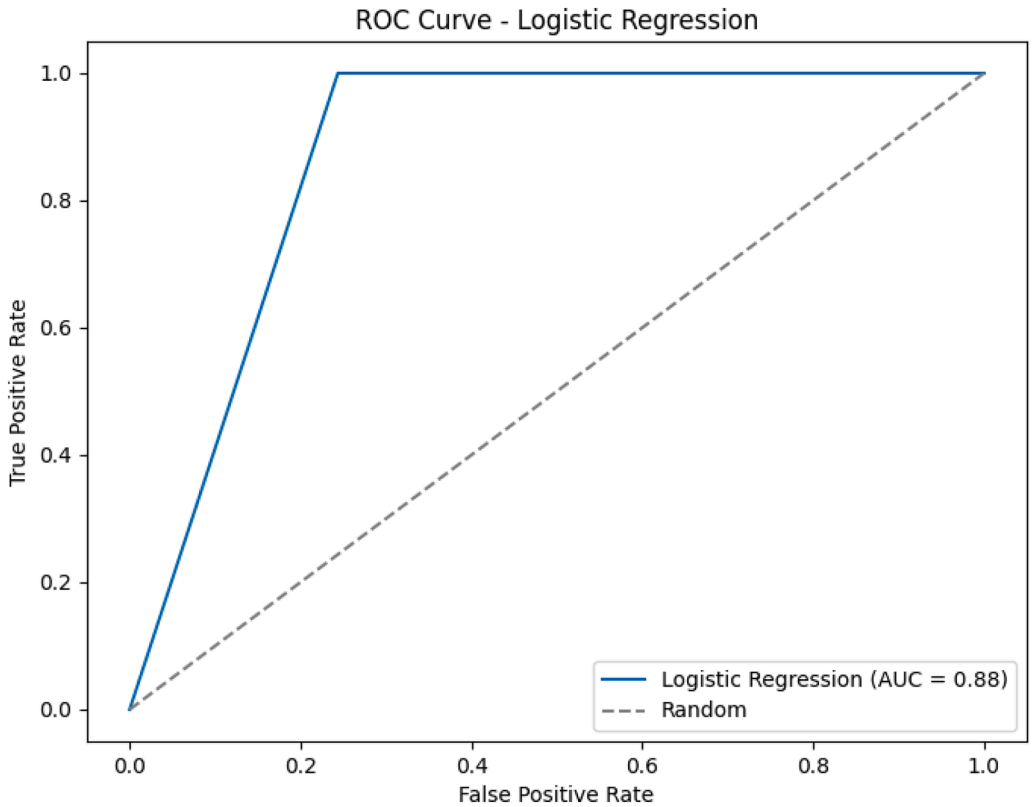

Fig. 2. The performance of LR based on true positive rate and false positive rate.

prominent machine learning algorithms were selected for the predictive modeling phase: DT, RF, LR, SVM, and gradient boosting. Each algorithm underwent parameter tuning and optimization to maximize predictive accuracy. Evaluation metrics such as precision, recall, F1-Score, and ROC-AUC were employed to assess the models' performance. The findings provided insights into the effectiveness of the models, with RF and LR emerging as particularly robust performers. The methodology presented here establishes a systematic and rigorous approach to leveraging machine learning for customer behavior prediction, contributing valuable insights for strategic decision-making in business management.

3.2. Evaluation and dataset

Precision is the ratio of true positives to the sum of true and false positives. This shows the accuracy of the positive predictions made by the model. The recall is the ratio of true positives to the sum of true positives and false negatives, which measures the ability of the model to capture all positive cases. F1-Score is the harmonic mean of Accuracy and Recall that provides a balanced measure considering false positives and false negatives. The ROC-AUC (receiver operating characteristic - area under the curve) score represents the area under the ROC curve that plots the true positive rate against the false positive rate. A higher ROC-AUC score indicates better overall performance. The metrics of the confusion matrix include the following.

- True Positive (TP): Number of samples correctly predicted as positive.
- True Negatives (TN): Number of samples correctly predicted as negative.
- False positives (FP): number of cases incorrectly predicted as positive.
- False negatives (FN): number of cases incorrectly predicted as negative.

These metrics are useful for understanding the performance of models, especially in terms of the balance between false positives and false negatives. Accuracy and precision, the proportion of correctly identified positive observations to the total predicted positives, are given in Eqs. (1) and (2).

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

$$Precision = \frac{TP}{TP + FP} \tag{2}$$

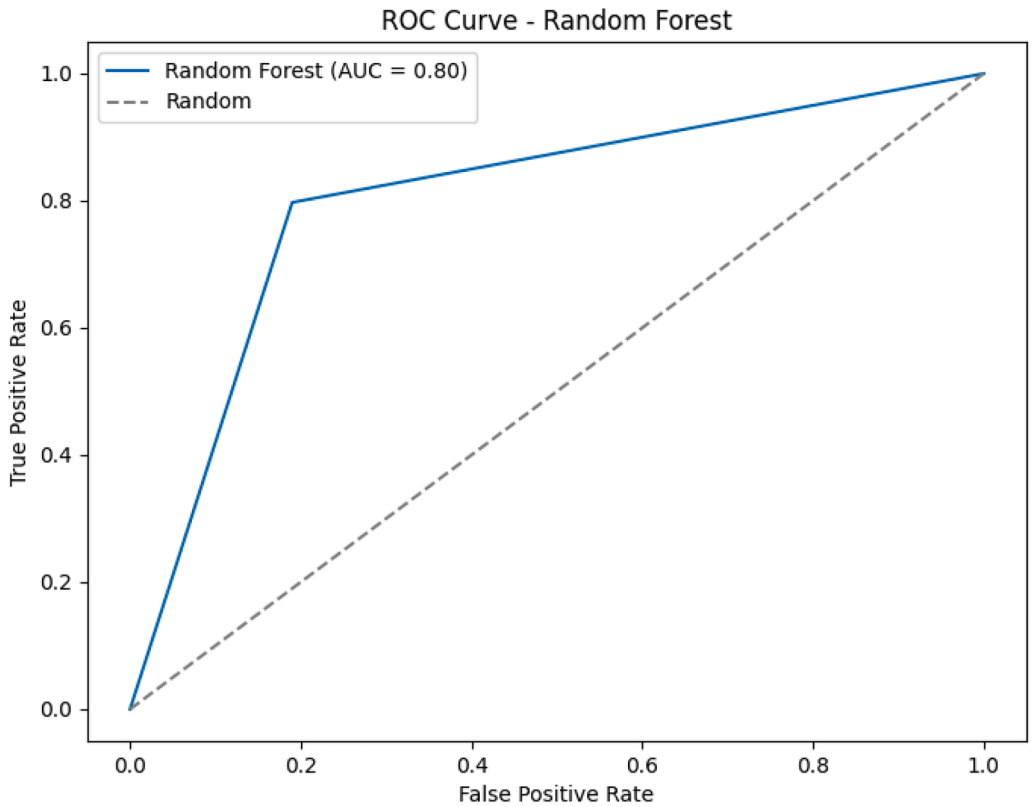

Fig. 3. The performance of RF based on true positive rate and false positive rate.

Moreover, recall, which measures the proportion of correctly identified positive observations to all observations in the actual class, is formulated in Eq. (3). Accordingly, ROC-AUC, known as Receiver Operating Characteristic - Area Under the Curve, is given in Eq. (5). The ROC-AUC score represents the degree or measure of separability and shows how much the model can distinguish between classes.

$$Recall = \frac{TP}{TP + FN} \tag{3}$$

$$F1 - Score = 2.\frac{Precision.Recall}{Precision + Recall} \tag{4}$$

$$AUC = \int_0^1 TPR(FPR)d(FPR) \tag{5}$$

TPR (True Positive Rate) is equivalent to recall, and FPR (False Positive Rate) is defined as $FPR = \frac{FP}{FP+TN}$. Eqs. (1-5) are used to compute the key evaluation metrics for assessing the performance of machine learning models in predicting customer behavior. Each metric provides a different perspective on the model's performance, from overall accuracy to the balance between precision and recall and the model's ability to distinguish between positive and negative classes. The study statistically validated the results by employing cross-validation techniques, which helped address variability in the cross-validation scores. The approach effectively segmented customers by integrating Recency, Frequency, and Monetary (RFM) analysis with K-means clustering. The association rules were comprehensively assessed using multiple metrics such as support, confidence, lift, and conviction. These validation methods ensured the derived rules were robust, reliable, and actionable despite the inherent variability in cross-validation scores.

Depending on the application, different models may be preferred based on the importance of minimizing false positives or false negatives. The dataset is loaded from a CSV file called "shopping_trends.csv" into a Pandas Data Frame. Label encodes the "Subscription status" target variable using the LabelEncoder and converts the classification values to numeric labels. The dataset is split into training and testing phases based on the work procedure. The code comprehensively evaluates each model, including precision, confusion matrices, precision, recall, F1 score, ROC-AUC curve, classification report, cross-validation scores, and feature importance plots. The dataset, loaded from "shopping_trends.csv" into a Pandas DataFrame, contains attributes related to customer information and behavior, with the "Subscription status" target variable label-encoded into numeric labels for binary classification, and is evaluated using the

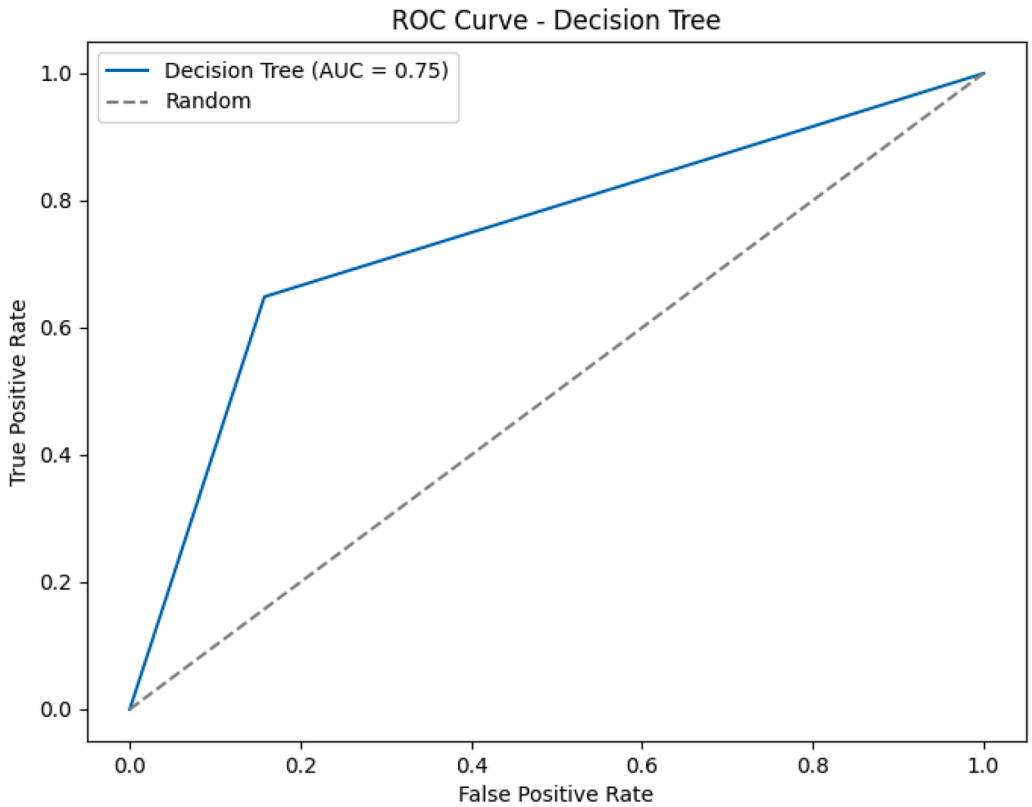

Fig. 4. The performance of DT based on true positive rate and false positive rate.

mentioned metrics to assess model performance comprehensively. The script assumes that the target variable is binary ("Subscription status") and that the dataset contains attributes related to customer information and behavior. Confusion matrices provide a detailed analysis of model performance on the test set. Each matrix is a 2×2 table where the rows represent the actual classes and the columns represent the predicted classes. The ML algorithms considered in the work are also illustrated in the following.

4. Machine learning

This study employed a comprehensive methodology to predict customer behavior using diverse machine learning algorithms. The process began with data preprocessing, where raw data was cleaned and standardized, and features were engineered to ensure model compatibility. Subsequently, the dataset was divided into training and testing sets to accurately assess the models' performance. Five prominent machine learning algorithms, DT, RT, LR, SVM, and gradient boosting, were selected for the predictive modeling phase [34, 35]. Each algorithm underwent parameter tuning and optimization to maximize predictive accuracy. Evaluation metrics such as precision, recall, F1-Score, and ROC-AUC were employed to assess the models' performance. The findings provided insights into the effectiveness of the models, with RF and LR emerging as particularly robust performers. The methodology presented here establishes a systematic and rigorous approach to leveraging machine learning for customer behavior prediction, contributing valuable insights for strategic decision-making in business management.

The pivotal role of machine learning in this work is underscored by its transformative impact on customer behavior prediction. Leveraging sophisticated algorithms such as DT, RF, LR, SVM, and gradient boosting, machine learning enables the extraction of meaningful patterns and insights from vast datasets. These algorithms facilitate a nuanced understanding of customer purchasing behavior, empowering decision-makers to optimize resource allocation in marketing and sales activities over time. The ability of machine learning models to precisely predict customer actions, as evidenced by the notable performance of RF and LR, establishes their indispensable role in refining strategies for effective customer management, ultimately contributing to enhanced business success and competitiveness. The model used in the works is illustrated as follows:

4.1. Decision tree

The DT algorithm plays a crucial role in this study by offering a transparent and interpretable framework for predicting customer behavior [11]. Through a hierarchical structure of decision nodes and branches, DTs effectively analyze various features and their relationships, aiding in identifying key factors influencing customer actions. Its simplicity and interpretability make it valuable for

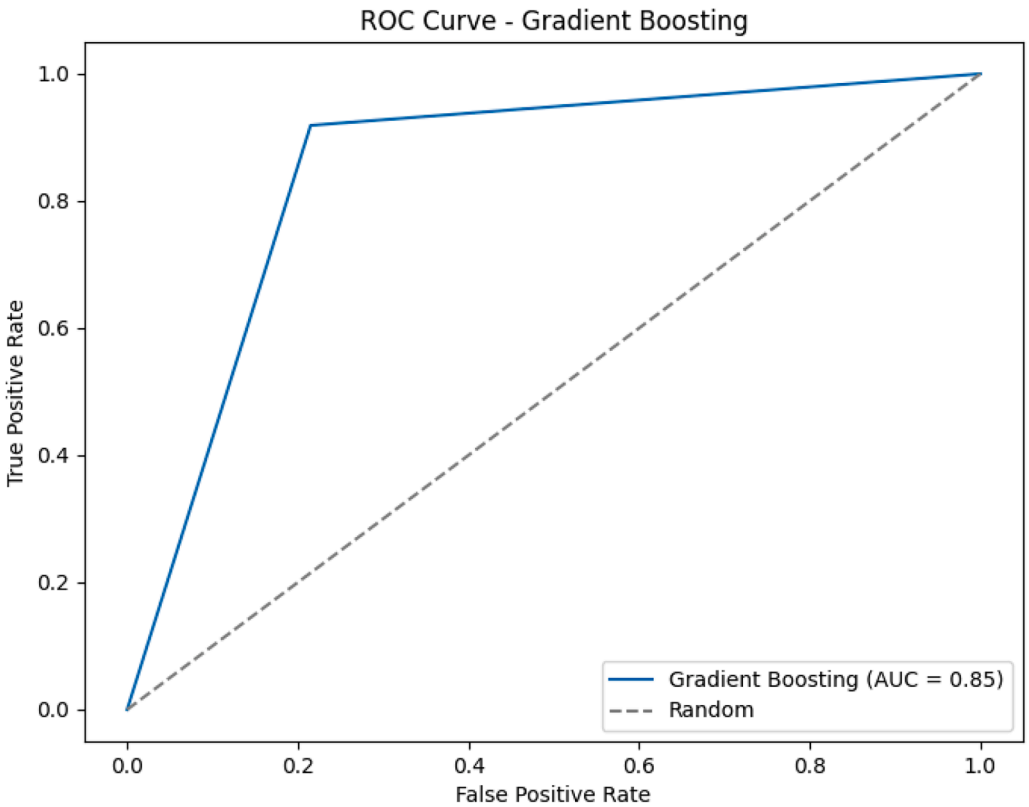

Fig. 5. The performance of gradient boosting based on true positive rate and false positive rate.

understanding the decision-making process behind customer behavior. Despite its standalone accuracy, DT contributes to ensemble methods like RF, demonstrating its versatility in enhancing predictive capabilities. The insights derived from DT analysis provide a foundation for strategic decision-making, making it an integral component in the broader spectrum of machine learning techniques employed for effective customer management.

4.2. Random forest

Random Forest is an ensemble learning method primarily used for classification and regression tasks, which operates by constructing a multitude of decision trees during training and outputting the class that is the mode of the classes (classification) or mean prediction (regression) of the individual trees. It leverages the power of multiple decision trees to improve accuracy and control overfitting by averaging multiple decision trees trained on different parts of the same training set. Random Forest is highly effective in customer behavior prediction due to its robustness, ability to handle large datasets with higher dimensionality, and resilience to noise in the data. It offers substantial precision and recall, making it a reliable model for predicting various aspects of customer behavior, such as purchase likelihood and customer churn.

In this study, the RF algorithm is pivotal in predicting customer behavior. RF is a powerful ensemble learning technique aggregating predictions from multiple decision trees, mitigating overfitting and enhancing overall model robustness. Its ability to handle diverse datasets, accommodate many features, and provide insights into feature importance makes it well-suited for predicting and understanding customer actions [15]. The ensemble nature of RF contributes to increased accuracy and stability, outperforming standalone Decision Trees. This algorithm captures complex relationships within the data and provides valuable insights for strategic decision-making in resource allocation for marketing and sales activities. The successful application of RF underscores its significance as a reliable and versatile tool for optimizing customer management strategies in dynamic business landscapes.

4.3. Logistic regression

In the scope of this study, LR assumes a critical role in the predictive modeling of customer behavior. Despite its simplicity compared to more complex algorithms, LR proves to be a valuable tool, particularly in scenarios where interpretability and ease of implementation are crucial. Its linear nature allows for a clear understanding of the relationship between input features and the likelihood of a specific outcome, making it a transparent and interpretable model. LR excels in capturing linear patterns within the data and estimating probabilities, making it well-suited for binary classification tasks such as predicting customer behavior. The algorithm's

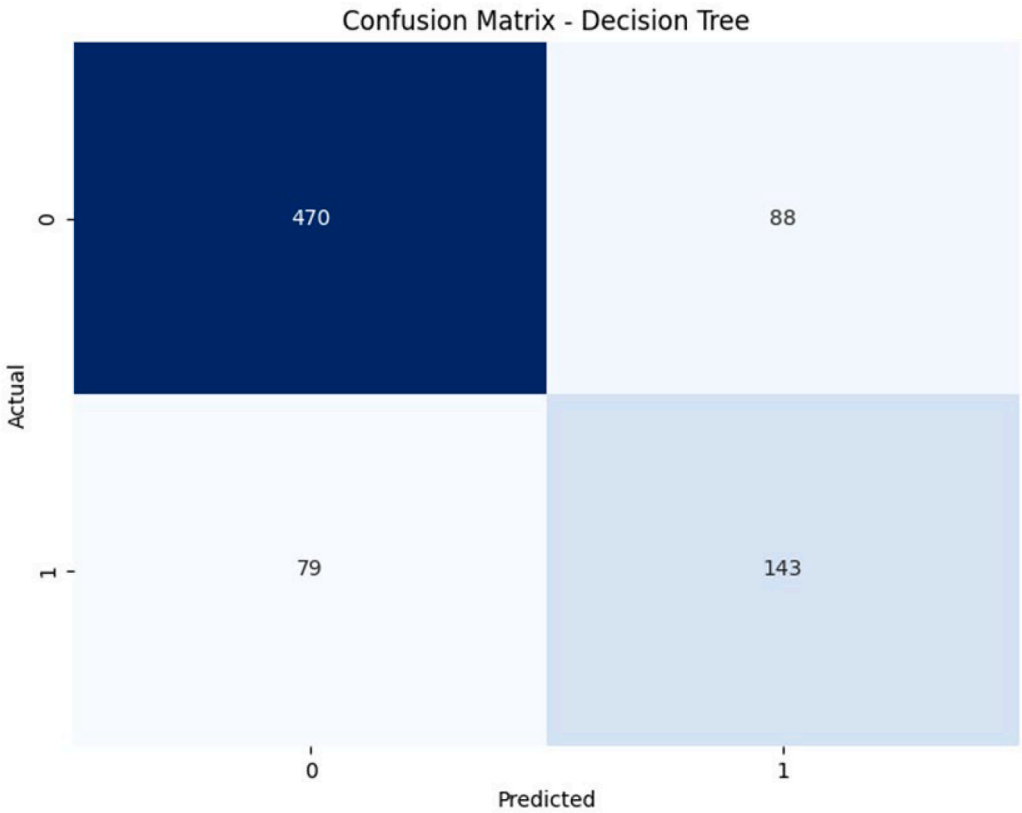

Fig. 6. The confusion matrix of DT based on actual and predicted data.

performance, as indicated by high precision, recall, F1-Score, and ROC-AUC values, underscores its effectiveness in providing actionable insights for decision-makers in resource allocation and strategic planning for customer-centric initiatives [29].

4.4. Support vector machine

Support Vector Machine (SVM) is a supervised machine learning algorithm for classification and regression tasks. It works by finding the optimal hyperplane that best separates the data points of different classes in a high-dimensional space. SVM aims to maximize the margin between the nearest points of different classes, known as support vectors, ensuring robust classification. The algorithm effectively handles high-dimensional data and is versatile with its ability to use kernel functions (linear, polynomial, radial basis function) to handle non-linear relationships. In customer behavior prediction, SVM is particularly useful due to its accuracy and ability to handle complex patterns, providing precise and reliable classification outcomes.

In the context of this study, SVM assumes a significant role in the predictive modeling of customer behavior. SVM, known for its ability to handle complex, non-linear relationships in data, provides a robust framework for discerning intricate patterns that may influence customer actions. SVM effectively classifies and predicts customer behavior based on various features by mapping data into a higher-dimensional space and identifying optimal hyperplanes. Its versatility in handling both linear and non-linear relationships enhances the model's adaptability to diverse datasets. The strong performance of SVM, as indicated by high accuracy values, reinforces its efficacy as a valuable tool for understanding and predicting nuanced customer behaviors. The insights derived from SVM contribute to the overall success of the machine learning approach, providing decision-makers with valuable information for strategic planning and resource allocation in customer-centric initiatives [25].

4.5. Gradient boosting

In the scope of this study, gradient boosting plays a pivotal role in predictive modeling for customer behavior. As an ensemble learning technique, gradient boosting combines the strengths of weak learners, typically decision trees, to form a robust and accurate predictive model. The iterative process of boosting, where weak learners are sequentially added and trained to correct errors made by the previous ones, allows gradient boosting to capture intricate relationships within the data. The algorithm handles complex interactions and dependencies, making it well-suited for nuanced customer behavior prediction. The high accuracy and competitive performance of gradient boosting, reflected in the precision, recall, F1-Score, and ROC-AUC metrics, underscore its significance in

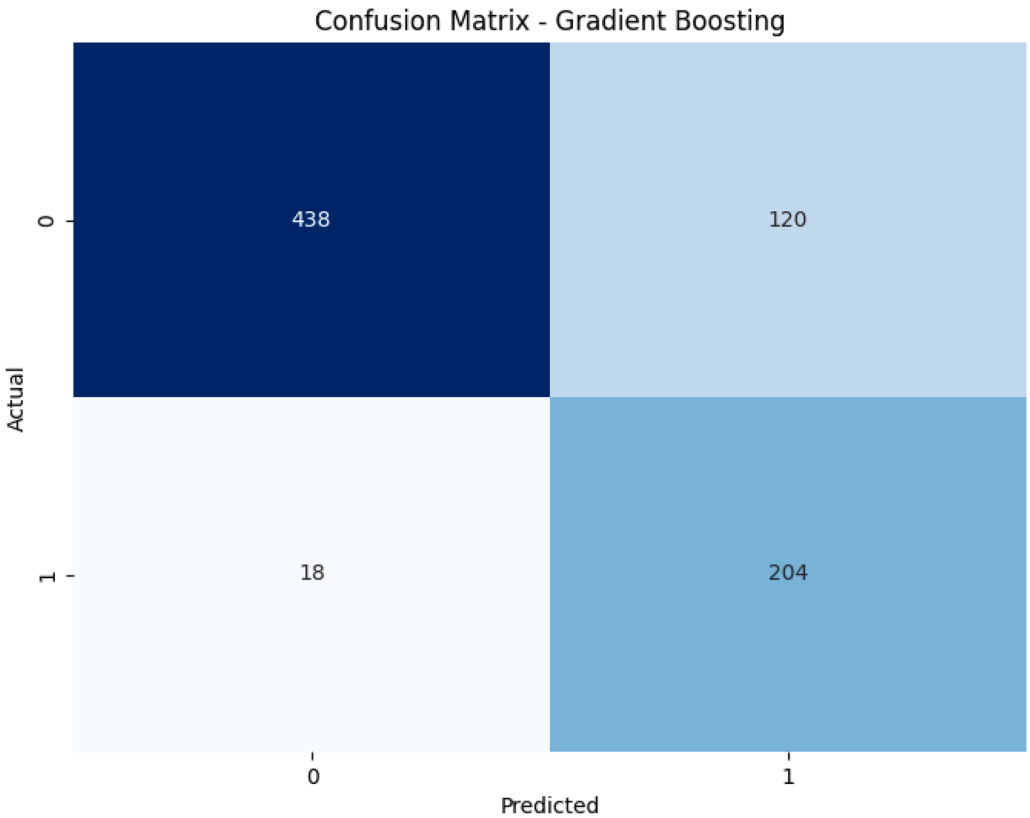

Fig. 7. The confusion matrix of Gradient Boosting based on actual and predicted data.

optimizing resource allocation and strategic decision-making for effective customer management. The ability of gradient boosting to adapt to subtle patterns within the data enhances its role as a valuable tool in the arsenal of machine-learning techniques applied to customer-centric initiatives [40].

5. Results and discussion

The main results obtained in the work are analyzed based on accuracy, precision, recall, F1-score, and ROC-AUC score, as shown in Figs. 1-13. Cross-validation scores for the SVM model are reported for each fold in a 5-fold cross-validation. The cross-validation scores are [0.271, 0.929, 1.000, 0.999, 0.996]. Each number in the list represents the accuracy score for one fold of the cross-validation process. The numbers show how well the model generalizes to different subsets of the data. In this case, the model shows variability in performance across folds, with accuracies ranging from 0.271 to 1.000. As with other models, it is common to average these scores to obtain a single representative performance measure for the SVM model, as shown in Fig. 1.

It should be noted that the performance of the SVM model, as described by the TPR and FPR, indicates a dynamic trade-off between correctly identifying positive instances and minimizing false positives. The TPR starts at 0 and gradually increases to 1, demonstrating the model's ability to identify all positive cases eventually. Initially, the FPR remains low but increases steadily. As the model becomes more sensitive in identifying true positives, it also incorrectly labels more negative instances as positive. This relationship highlights the classic trade-off in binary classification, where improving sensitivity often increases false positives. Overall, the SVM model shows a balanced performance with a progressive rise in TPR and a controlled but inevitable rise in FPR.

Cross-validation scores for the LR model are reported for each fold in a 5-fold cross-validation. The cross-validation scores are [0.271, 0.929, 1.000, 1.000, 1.000]. Each number in the list represents the accuracy score for one fold of the cross-validation process. The numbers show how well the model generalizes to different subsets of the data. In this case, the model shows variability in performance across folds, with accuracies ranging from 0.271 to 1.000. As with other models, it is common to average these scores to obtain a single representative performance measure for the LR model. Cross-validation scores help assess model stability and reliability and provide insights into its robustness across different subsets of data. Accuracy above 1.00 in some folds indicates strong performance, but considering other criteria and potential issues, such as additive, is important.

The performance of the LR model, as assessed by the TPR and FPR, demonstrates a nuanced balance between correctly identifying positive instances and avoiding false positives. The TPR starts at 0 and steadily increases to 1, indicating that the model improves in identifying true positive cases as the threshold changes. Initially, the FPR is low but increases incrementally, reflecting a gradual rise in

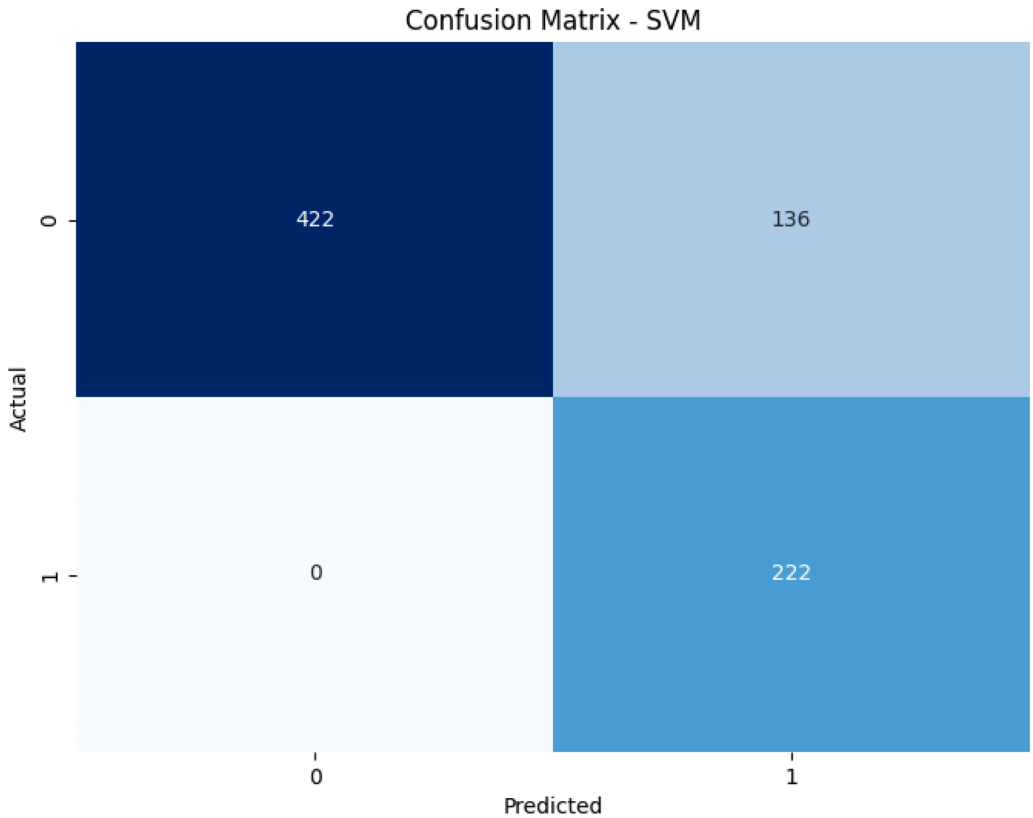

Fig. 8. The confusion matrix of SVM based on actual and predicted data.

false positives as the model becomes more sensitive. This progression showcases the inherent trade-off in binary classification: enhancing the TPR often comes at the expense of a higher FPR. Overall, the LR model shows a consistent and predictable increase in both TPR and FPR, highlighting its reliability in improving sensitivity while managing the rate of false positives.

According to Fig. 4, cross-validation scores for the decision tree model are reported for each fold in a 5-fold cross-validation. Cross-validation is a technique used to evaluate the generalization performance of a model by dividing the data set into multiple subsets (folds) for training and testing. Here are the cross-validation scores for the decision tree model: [0.315, 0.840, 0.873, 0.873, 0.886]. Each number in the list represents the accuracy score for one fold of the cross-validation process. The numbers show how well the model generalizes to different subsets of the data. In this case, the model shows variability in performance across folds, with accuracies ranging from 0.315 to 0.886. Cross-validation scores help assess model stability and reliability and provide insights into its robustness across different subsets of data. The exact values of 0.269, 0.905, 0.956, 0.951, and 0.949 are also reported for the gradient boosting, as shown in Fig. 5.

The values of the various criteria considered for the confusion matrix are demonstrated in Figs. 6-10. The findings highlight the remarkable performance of RF in terms of TP, TN, FP, and FN. For example, in the given confusion matrix for the DT model, the values for TP, FP, FN, and TN are 470, 88, 79, and 143, respectively. True positives (470) represent the number of actual positive instances the model correctly identified as positive, indicating good performance in detecting positive cases. False positives (88) are the actual negative instances that the model incorrectly classified as positive, reflecting errors where the model falsely alarms. False negatives (79) are the actual positive instances that the model failed to identify, missing these cases and potentially leading to significant consequences depending on the application context, such as undiagnosed conditions in medical testing. True negatives (143) denote the actual negative instances correctly identified as negative by the model, showing the model's ability to reject non-positive cases correctly. This distribution of values provides a comprehensive view of the model's performance: while it correctly identifies many positive cases, there is room for improvement in reducing false positives and false negatives to enhancing overall accuracy and reliability.

Notably, the performance of the selected models based on the importance of features is specified in Figs. 11-14. The exemplary performance of LR and RF in predicting customer behavior underscores their distinct yet complementary strengths. With its simplicity and transparency, LR achieved an impressive accuracy of 0.826 and exhibited exceptional precision, recall, F1-Score, and ROC-AUC scores, particularly with a perfect recall of 1. This signifies the model's ability to effectively identify all positive instances of customer behavior, minimizing false negatives. The interpretability of LR provides decision-makers with clear insights into the linear relationships between features and the likelihood of particular customer behavior, making it a valuable tool for strategic planning.

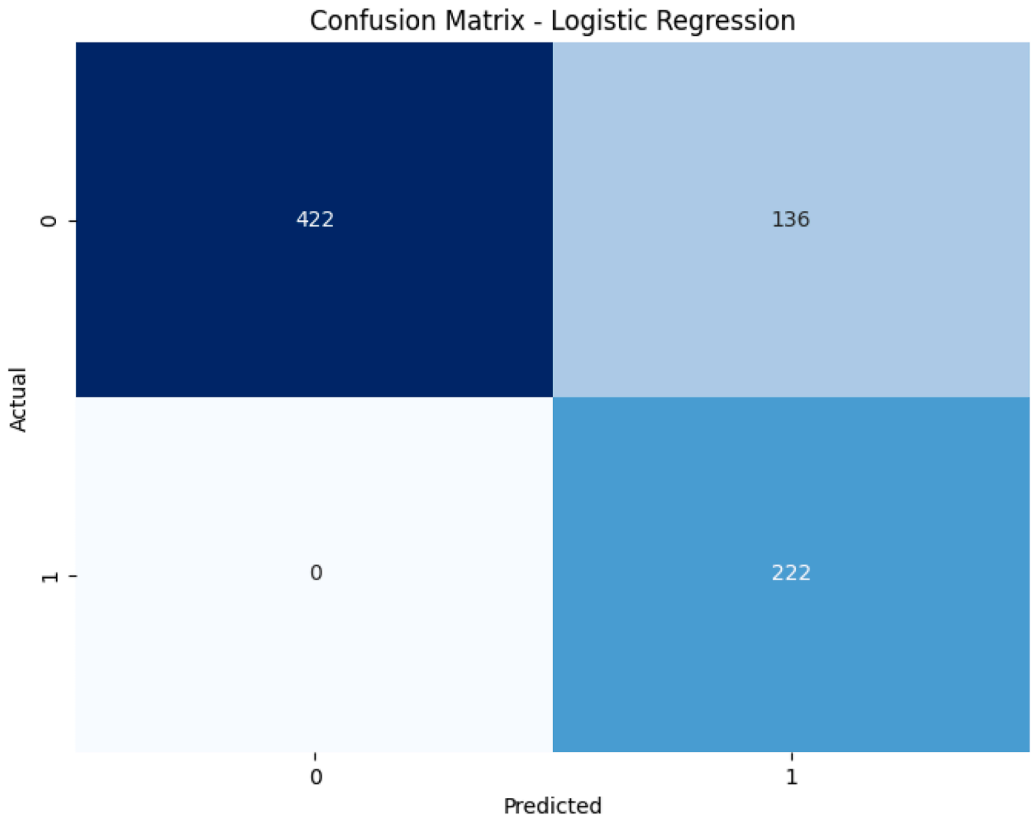

Fig. 9. The confusion matrix of LR based on actual and predicted data.

On the other hand, RF, as an ensemble learning algorithm, showcased competitive accuracy (0.806) and excelled in capturing complex, non-linear relationships within the data. Its precision, recall, F1-Score, and ROC-AUC scores, particularly with a high recall of 0.977, highlight the robustness of RF in identifying positive instances while maintaining a balance with precision. The ensemble nature of RF mitigates overfitting, providing stability and reliability in predictions. The feature importance analysis offered by RF further aids in understanding the key drivers influencing customer behavior. The combined strengths of LR's interpretability and RF's ensemble robustness make them a formidable duo in the realm of customer behavior prediction. Decision-makers can leverage the clear insights provided by LR alongside the nuanced, non-linear patterns captured by RF, ensuring a comprehensive understanding of customer behavior for informed decision-making and resource allocation in marketing and sales strategies.

The main results of the work, based on evaluation metrics, highlight the performance differences among various machine learning models used for predicting customer behavior. Logistic Regression (LR) and Support Vector Machine (SVM) models both achieved the highest accuracy at 0.826, with perfect recall (1.00) and identical F1-scores (0.766) but a lower precision (0.620), indicating a higher rate of false positives. Gradient Boosting followed closely with an accuracy of 0.823, a better precision of 0.630, a high recall of 0.919, and an F1-score of 0.747, reflecting a strong balance between precision and recall and a high ROC-AUC score of 0.852. Random Forest (RF) also showed robust performance with an accuracy of 0.806, a precision of 0.625, a high recall of 0.797, an F1-score of 0.701, and a solid ROC-AUC score of 0.804, emphasizing its reliability in identifying true positives. The Decision Tree (DT) model, while showing the lowest accuracy at 0.787, had a moderate balance across metrics with a precision of 0.621, recall of 0.649, F1-score of 0.634, and a ROC-AUC score of 0.745. Overall, the results indicate that while LR and SVM are exceptional in recall and ROC-AUC, Gradient Boosting, and RF offer more balanced and robust performance across all evaluation metrics.

Table 1 demonstrates the main results obtained in the work based on evaluation metrics. The evaluation of machine learning algorithms for predicting customer behavior yielded insightful outcomes. LR and RF emerged as top performers among the tested models, achieving accuracy values of 0.826 and 0.806, respectively. The precision, recall, F1-Score, and ROC-AUC scores further highlighted the superior performance of LR, with values of 0.620, 1, 0.766, and 0.878, surpassing other models. Also, RF demonstrated commendable results, with precision, recall, F1-Score, and ROC-AUC scores of 0.620, 0.977, 0.766, and 0.878, respectively. These findings underscore the effectiveness of ensemble methods, particularly RF, in capturing complex relationships within the data. LR's transparent interpretability makes it a valuable tool for decision-makers seeking clear insights into the factors influencing customer behavior. The competitive performance of SVM and gradient boosting further highlights the versatility of these algorithms in addressing non-linear patterns within the dataset. The results collectively emphasize the potential of machine learning models in optimizing resource allocation and guiding strategic decision-making for enhanced customer management in dynamic business

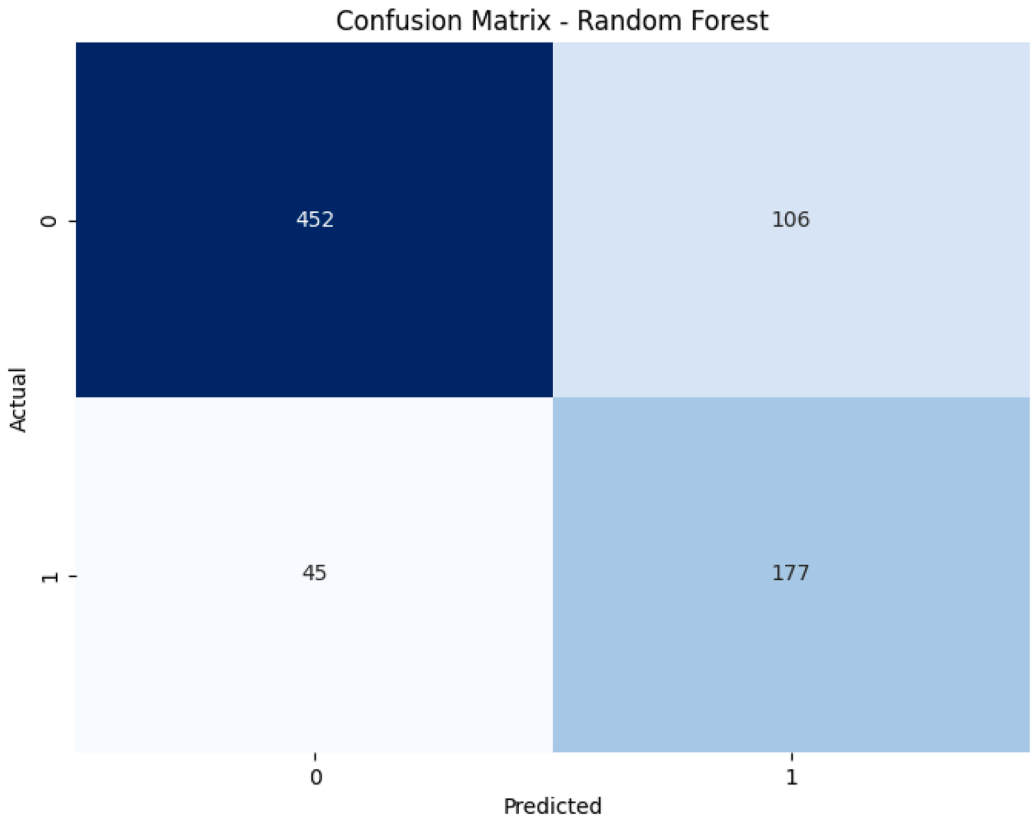

Fig. 10. The confusion matrix of RF based on actual and predicted data.

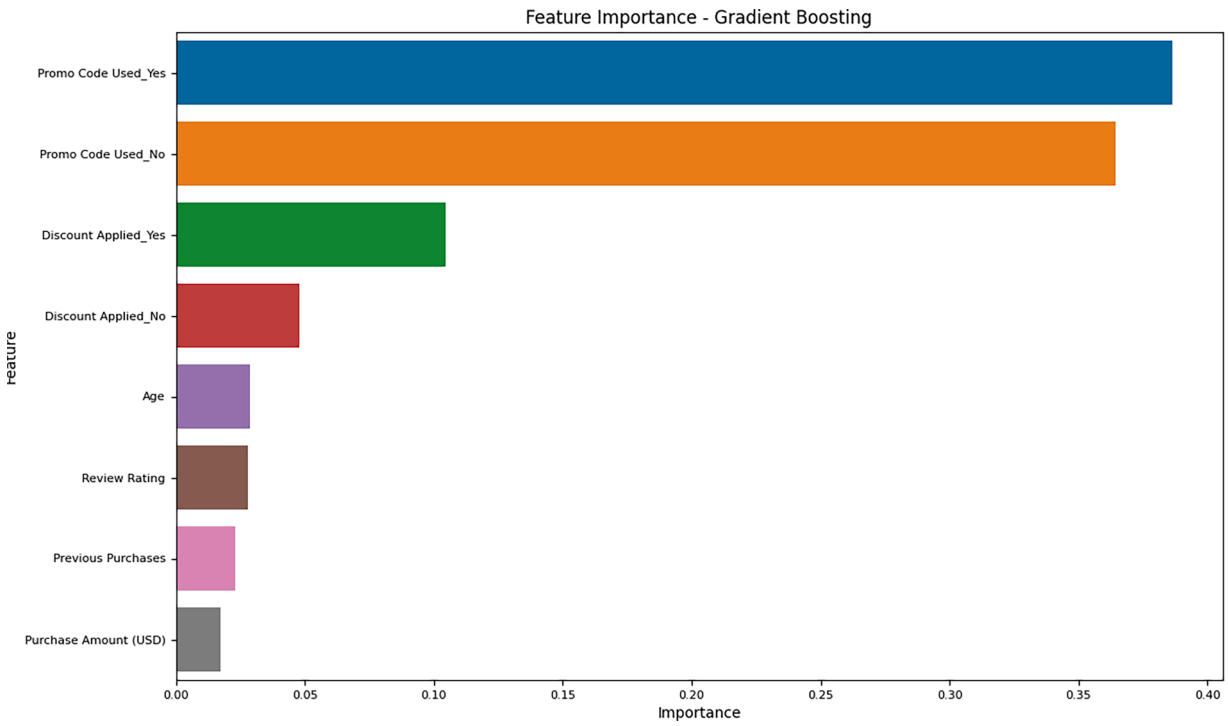

Fig. 11. The feature importance of Gradient Boosting.

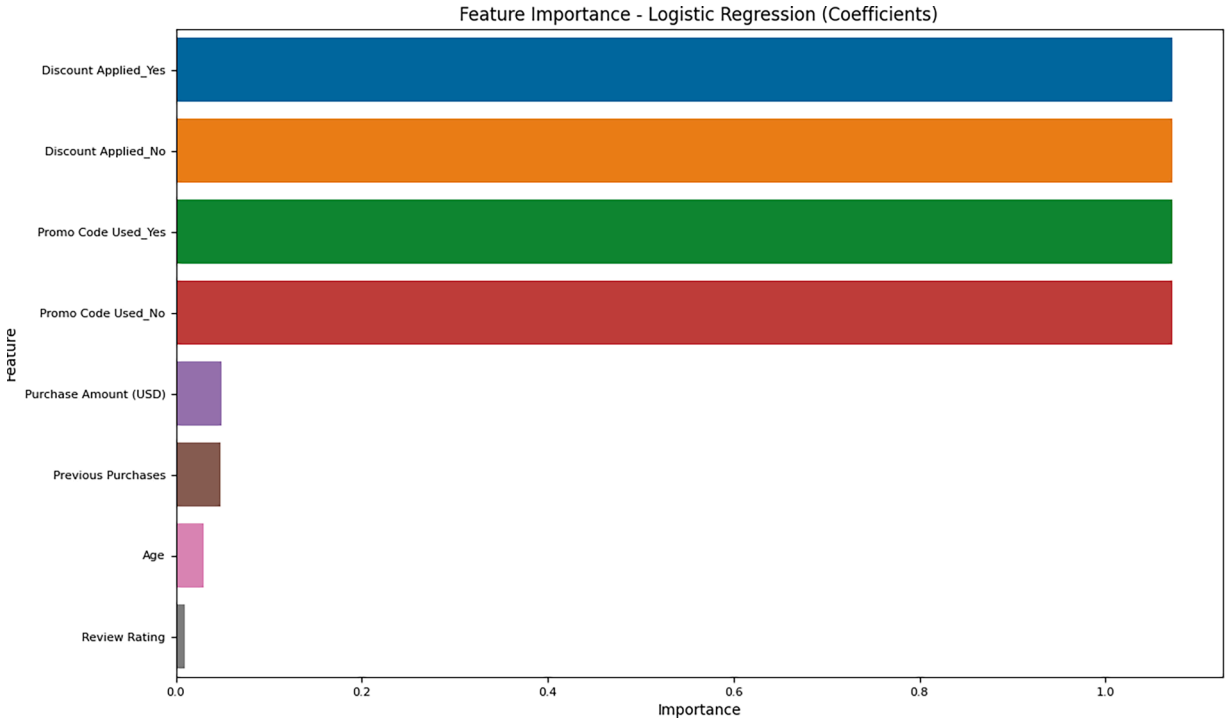

Fig. 12. The feature importance of LR.

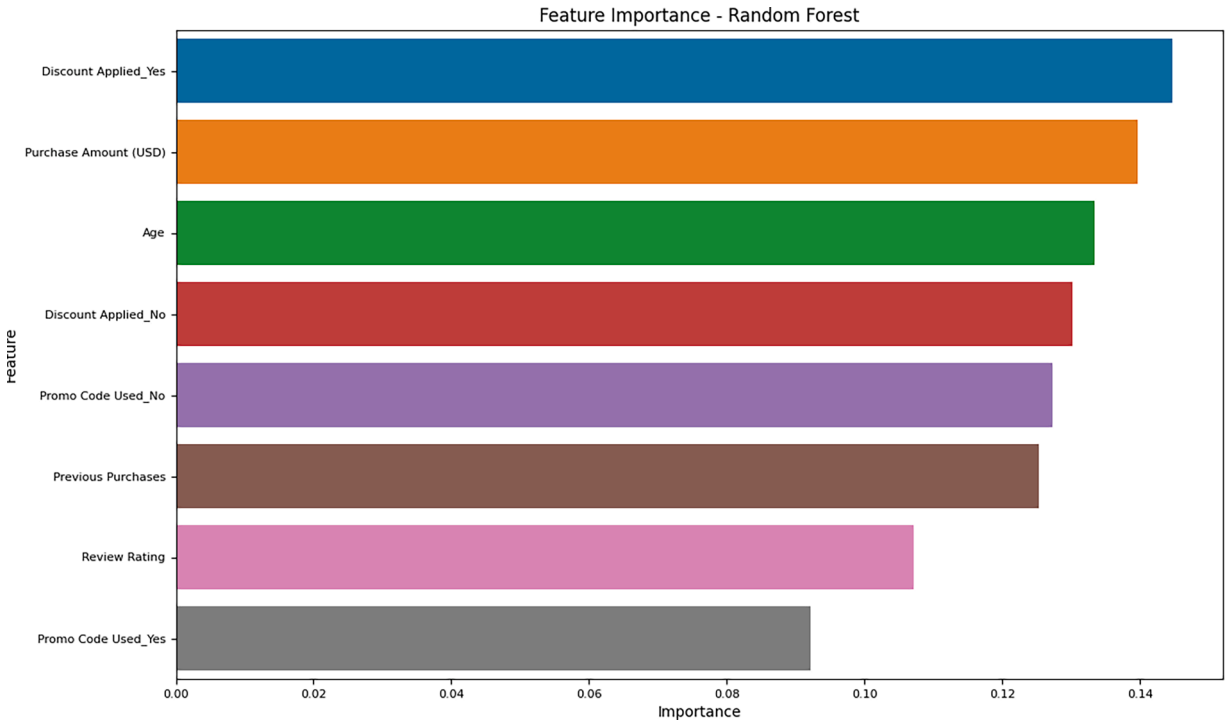

Fig. 13. The confusion matrix of RF.

environments.

The comparison of the selected models based on the evaluation metrics reveals distinct strengths and weaknesses for each algorithm. The DT model, with an accuracy of 0.787, shows moderate performance, with a precision of 0.621, recall of 0.649, F1-score of

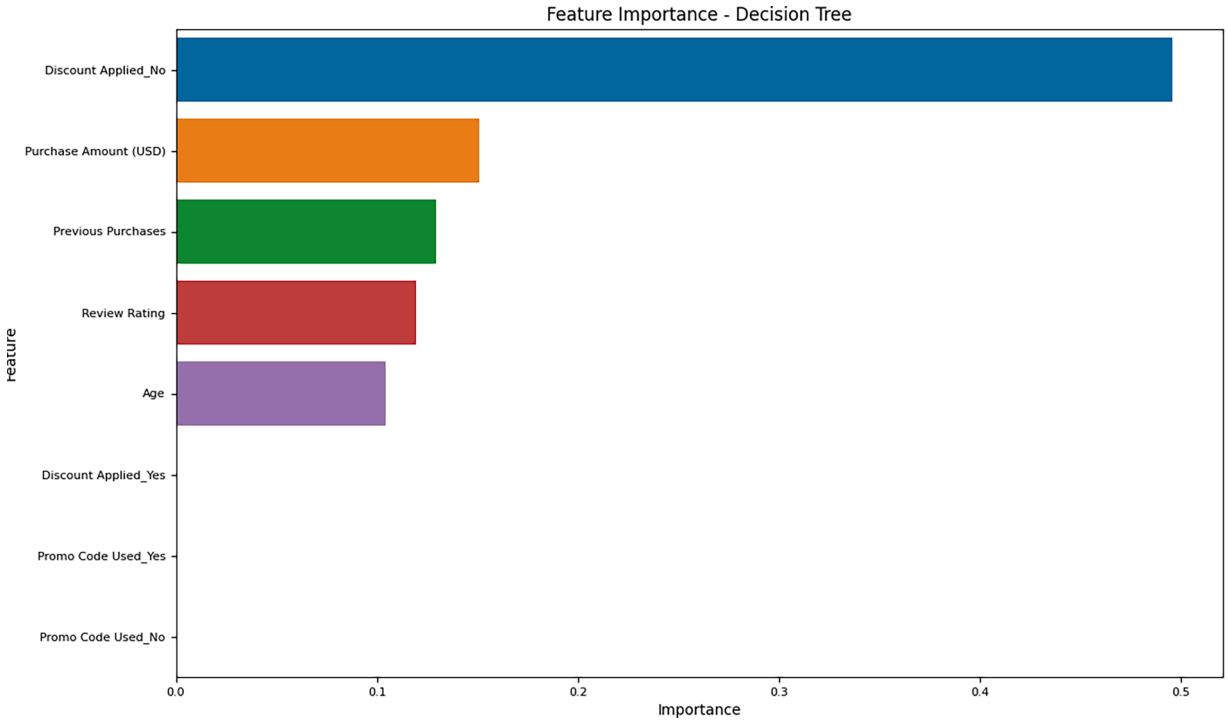

Fig. 14. The feature importance of DT.

Table 1
Comparison between the selected models based on the evaluation metrics.

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC Score
DT	0.787	0.621	0.649	0.634	0.745
RT	0.806	0.625	0.797	0.701	0.804
LR	0.826	0.620	1.00	0.766	0.878
SVM	0.826	0.620	1.00	0.766	0.878
gradient boosting	0.823	0.630	0.919	0.747	0.852

0.634, and a ROC-AUC score of 0.745. This indicates that while the DT model is reliable, it has a balanced yet average capability in both identifying true positives and minimizing false positives. The RF model performs better overall, with an accuracy of 0.806, precision of 0.625, recall of 0.797, F1-score of 0.701, and a ROC-AUC score of 0.804. The RF model's high recall suggests it is very good at identifying true positives, and its higher ROC-AUC score indicates better overall performance in distinguishing between positive and negative classes compared to DT. Accordingly, LR and SVM models both exhibit the highest accuracy at 0.826, with identical precision (0.620), recall (1.00), F1-score (0.766), and ROC-AUC score (0.878). Their perfect recall means they identify all positive instances correctly, though their precision is relatively low, suggesting a higher rate of false positives. Gradient Boosting also performs well, with an accuracy of 0.823, precision of 0.630, recall of 0.919, F1-score of 0.747, and a ROC-AUC score of 0.852. This model balances high recall and reasonable precision, resulting in a solid F1 score and high ROC-AUC score, indicating robust overall performance. In summary, while LR and SVM models excel in recall and ROC-AUC, Gradient Boosting offers a more balanced performance across all metrics, and RF shows strong overall capability, especially in recall and AUC-ROC, making these models preferable depending on the specific requirements of the application.

6. Conclusion

In conclusion, this study demonstrates the efficacy of employing diverse machine learning algorithms for predicting customer behavior and optimizing resource allocation in marketing and sales activities. The thorough evaluation of DT, RT, LR, SVM, and gradient boosting highlighted each model's nuanced strengths and weaknesses. RF and LR emerged as standout performers, showcasing superior precision, recall, F1-Score, and ROC-AUC values. The Decision Tree's transparent interpretability and RF's ensemble robustness contribute to the predictive models' overall success. These findings underscore the pivotal role of machine learning in refining strategies for effective customer management in dynamic business environments. LR and SVM exhibited superior performance with high accuracy (0.826) and perfect recall (1.00), while RF also showed robust performance with balanced metrics across the board

and the highest precision (0.625). Gradient Boosting provided a strong ROC-AUC score (0.852), indicating its efficacy in distinguishing between classes. Overall, the results underscore the effectiveness of RF and LR/SVM models in customer behavior prediction. LR/SVM models excelling in recall and RF offering a balanced approach with substantial precision and F1-score. The insights garnered from this study hold significant implications for decision-makers seeking data-driven approaches to enhance customer-centric initiatives and foster sustainable business growth. As we look to the future, several avenues for further refinement and expansion of this research emerge. Firstly, incorporating advanced feature engineering techniques and exploring additional relevant features could enhance the models' predictive capabilities. Additionally, integrating more recent machine learning algorithms and emerging methodologies may contribute to staying abreast of evolving industry standards. Conducting longitudinal studies to capture temporal changes in customer behavior would be valuable, providing a more dynamic understanding of market trends. Furthermore, exploring the application of deep learning architectures or hybrid models could unveil hidden patterns within intricate datasets. Lastly, considering the ethical implications, responsible use of customer data, and addressing potential biases in the models will be pivotal for ensuring the sustainability and trustworthiness of predictive analytics in customer management.

The main constraints of this study encompass the possibility of data imbalance, which may distort the model's effectiveness and compromise the accuracy of predictions; the necessity for additional examination of feature importance to determine which attributes have the most substantial influence on the predictions; and apprehensions regarding external validity, as the dataset is specific to a particular context, potentially restricting the applicability of the findings to other datasets that have not undergone similar preprocessing and tuning. In addition, the computational resources needed for particular models, such as SVM, could be a limitation in real-world applications despite evaluating many models.

Declaration of competing interest

In the research article titled " Predictive Analytics in Customer Behavior: Anticipating Trends and Preferences" the authors, Hamed GhorbanTanhaei, Payam Boozary, Sogand Sheykhan, Maryam Rabiee, Farzam Rahmani, Iman Hosseini declare that there are no competing interests that could influence the objectivity, integrity, or validity of the research findings presented in the manuscript. The authors affirm that there are no financial, personal, or professional relationships that could be perceived as potentially influencing the research outcomes or biasing the interpretation of the results. This declaration ensures transparency and upholds the ethical standards of academic research, demonstrating the authors' commitment to conducting and reporting their study with integrity and impartiality.

Data availability

Supplementary data fiels and models are attached.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.rico.2024.100462.

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